Sentiment Analysis & Data Insights

Professional Project Documentation

This document provides technical and functional documentation for the AI-powered Sentiment Analysis & Data Insights web application. The project analyzes batches of user-generated text and produces sentiment classifications, visual analytics, and downloadable insights reports.

Project Type	AI-Powered Web Application
Technology Stack	React, TanStack Start, AI SDK, Gemini AI, Tailwind CSS
Deployment Support	Cloudflare Workers, Vercel, Netlify, Docker
Primary Function	Sentiment Analysis and Insights Generation

Project Overview

The Sentiment Analysis & Data Insights application is designed to process large batches of textual data such as reviews, tweets, and survey responses. The platform uses artificial intelligence to classify sentiment, identify emotional tone, extract keywords, and generate insight reports.

Key Features

• Sentiment classification for text inputs
• Confidence scoring and emotional analysis
• Keyword extraction from user content
• KPI dashboard with charts and visual reports
• Downloadable Markdown insights report
• Responsive and modern user interface

Technology Stack

The project uses a modern full-stack architecture:

Frontend: React with TanStack Start and Tailwind CSS v4
Backend: Edge server routes with AI SDK integration
AI Provider: Google Gemini Flash Preview via Lovable AI Gateway
Charts: Recharts for visual analytics and reporting

System Requirements

The following prerequisites are required before running the application:

• Bun installed locally
• Lovable account or self-hosting environment
• Environment variable: LOVABLE_API_KEY

Development and Deployment

The application supports multiple hosting environments including:

• Lovable Cloud

• Cloudflare Workers
• Vercel
• Netlify
• Docker / VPS deployments

Production builds are generated using:

```
bun run build
```

Environment Variable Configuration

Variable	Required	Description
LOVABLE_API_KEY	Yes	Lovable AI Gateway API key used for AI requests

Application Workflow

1. User submits text data into the application.
2. The AI model processes the text using Gemini Flash Preview.
3. Sentiment, confidence, emotions, and keywords are extracted.
4. Results are visualized through charts and KPI cards.
5. The user downloads the generated insights report.

Conclusion

The Sentiment Analysis & Data Insights project demonstrates the integration of artificial intelligence with modern web technologies to deliver actionable business insights from textual data. The application is scalable, deployment-ready, and optimized for real-time sentiment analytics.